When Is a User's Own History Enough? A Controlled Study of Peer-Isolated Personal Recommendation

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Abstract. Personal data analytics systems often cannot consult other users: the data may be on-device, legally siloed, or simply absent for a new deployment. We ask when recommendation built *exclusively* from a single user's own interaction history is sufficient, and when it is not. We use Essence—a deliberately minimal peer-isolated system pairing a frozen sentence encoder with per-user K-means and cosine retrieval—as an instrument rather than a proposal, evaluating it against nine baselines on three datasets spanning music, e-commerce and film. Two findings emerge. First, the answer is governed by the density of cross-user co-occurrence in the evaluated population, not by the content domain. Peer-isolated retrieval ranks second and third of ten where that density is low (Last.fm-1K, Amazon Books) and eighth of ten where it is high (MovieLens-25M); the pattern belongs to the whole no-pooling architecture family rather than to any particular component. We demonstrate the mechanism *within a single dataset*: raising Last.fm-1K's user cap lifts the share of items heard by two or more users from 10.3% to 28.4% and reverses the verdict, from $d = +0.28$ in favour of peer-isolated retrieval over item-based collaborative filtering to $d = -0.16$ against it, both significant. A domain is therefore not weak or strong in collaborative signal; a *cohort* is. Second, and centrally, the interest-clustering step contributes nothing measurable. In the targeted test on the users whose taste is most clearly multi-modal it is significant in none of 18 silhouette-stratified comparisons; and across the fourteen dataset-level comparisons spanning three datasets, two sentence encoders, two user-activity cohorts and two temporal-split protocols, exactly one favours clustering and one favours simple recency-weighting, both marginal. We report the single exception rather than suppress it, and show it falls below its own detection threshold. We also report a two-sided bootstrap defect, present in our earlier work, that counted tied resamples as evidence against the null and inflated significance on sparse long-tail metrics. Baselines throughout are given the benefit of the doubt, a convention that costs us one previously significant result.

Keywords: Personal data analytics · Recommender systems · Negative results · Cold start · Reproducibility

1 Introduction

A recommender that may not look at other users is an unusual object. Almost all deployed systems pool behaviour across a population, and the dominant methods—collaborative filtering, matrix factorisation, sequential models—are defined by that pooling. Yet a growing class of settings forbids it. On-device assistants hold data that never leaves the phone. Regulated domains silo records by subject. A new product has no population to pool over. In each case the practical question is the same: *how much is lost by modelling a user from their own history alone?*

This paper answers that question empirically and reports a negative result at its centre.[1]

We build a minimal peer-isolated system, Essence, and treat it as an instrument. Essence embeds items with a frozen sentence encoder, partitions each user's history into K interest clusters with K-means, selects the cluster nearest the user's recent activity, and retrieves by cosine similarity. No component is novel and none is intended to be; the system exists to isolate the variable of interest, which is the absence of cross-user information.

Contributions.

1. **A population-density result.** Peer-isolated retrieval is competitive where cross-user co-occurrence is sparse and far behind where it is dense. We identify the mediator by manipulating it *within* a dataset — twice, by two different means — rather than inferring it from a three-dataset contrast, and show the effect belongs to the no-pooling family rather than to clustering.
2. **A clustering null, established at scope.** Per-user interest clustering gives no measurable benefit over recency-weighted retrieval across three datasets, two encoders, two activity cohorts and two split protocols, with a minimum detectable effect for every comparison so the claim is bounded rather than asserted.
3. **A statistical correction.** We fix a tie-handling defect in the two-sided bootstrap p-value used in our earlier work, which reported some zero-versus-zero comparisons as maximally significant.
4. **Conservative reporting.** Baselines are reported at the better of their published and validation-tuned configurations, costing us one previously significant result.

[1] An earlier and substantially weaker version of this study appeared as a non-archival poster at the New In ML workshop, NeurIPS 2026. References to "our earlier work" throughout denote that version. The present paper supersedes it: the K-selection rule, the significance testing, the baseline configurations and the headline finding itself have all changed, and we document each change and the direction it moves our claims. Of the analyses reported here, the encoder replication, the heavy-user and expanded-cohort replications, the power analysis, the sequential baselines, the study-wide multiplicity correction, the seed-stability check and the cohort-density result are new.

2 Related Work

Multi-interest recommendation. MIND [1] and ComiRec [2] represent each user by several interest vectors rather than one, on the premise that a single embedding cannot capture heterogeneous taste. Both learn those interests from pooled interaction data. Essence adopts the multi-interest premise but removes the pooling, which is precisely what lets us ask whether the premise survives in isolation.

Sequential recommendation. GRU4Rec [3] and SASRec [4] model the order of interactions. They require many users' sequences to learn transition structure, and we include both as library implementations.

Content-based and cold-start methods. Content-only retrieval avoids the cold-start problem structurally by not requiring co-occurrence. This is a generic property of the approach rather than a property of any particular system, a distinction the literature does not always preserve and one we are careful to make.

Negative and reproducibility results. Several authors have found that carefully tuned simple baselines match or exceed neural recommenders [5,6]. Our clustering null is of that kind, with the difference that the method whose component we find inert is our own.

3 Method

Let user u have interaction history $H_u = (i_1, \ldots, i_n)$ in chronological order. Each item i carries text t_i (title, artist or author, genre or category, description). We embed with a frozen sentence encoder ϕ, so $v_i = \phi(t_i)/\|\phi(t_i)\|$. No interaction data enters ϕ.

Interest clusters. We run K-means on $\{v_i : i \in H_u\}$ to obtain centroids $c_1, \ldots, c_K$ for that user alone. If $|H_u| < K$ the user cannot be clustered and falls back to the mean-embedding method; we report the frequency of this.

Active cluster and retrieval. Let r_u be the mean of the last ten items' embeddings. We select $c^\star = \arg\min_k \|c_k - r_u\|$ and return the ten unseen catalogue items of highest cosine similarity to $c^\star$.

Selecting K. Our earlier work selected K with a stopping rule evaluated forward from a checkpoint at $K = 8$, which a reviewer correctly identified as anchoring. We replace it with a criterion that is anchoring-free by construction: choose the K maximising validation Recall@10 subject to (a) mean items per cluster ≥ 3 and (b) every user clusterable. Constraint (a) prevents degenerate singleton clusters; (b) prevents a configuration whose apparent quality is partly produced by fallback. Table 1 gives the outcome.

Table 1. Adopted K under the anchoring-free criterion. Both changes are forced by constraints the original rule did not enforce.

Dataset	Published K	Adopted K	Reason for change
Last.fm-1K	10	10	unchanged
Amazon Books	10	**8**	$K = 10$ gives 2.58 items per cluster, below the floor of 3
MovieLens-25M	15	**10**	at $K = 15$ only ∼90% of users could cluster

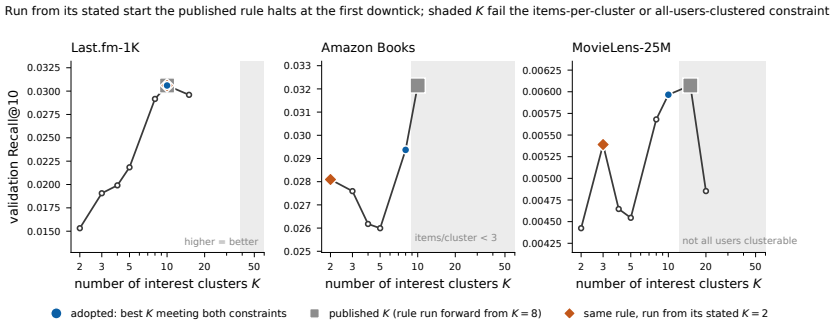


Fig. 1. Validation Recall@10 against K for each dataset, with grid points excluded by the occupancy floor or the coverage constraint marked. The adopted K is the constrained maximiser, not the unconstrained one: on Amazon Books and MovieLens-25M the unconstrained argmax violates a constraint the original rule did not enforce.

The adopted values are invariant to the sentence encoder. They are not, however, a property of the domain: on Amazon Books the same criterion selects $K = 8$ for users with ∼29 interactions and $K = 60$ for users with 330 (Sect. 5.4). Any single reported K is contingent on the user filter.

4 Experimental Setup

Datasets. Three domains, chosen to span a range of collaborative signal strength: music (Last.fm-1K), book retail (Amazon Books) and film (MovieLens-25M). Users are filtered to an activity band and, for the two larger sets, sampled; this gives 99, 2,000 and 2,000 users. Each history is split chronologically at that user's own 80th percentile. We also report a global-cutoff variant (Sect. 5.7).

Baselines. Nine: Recency-Weighted, Last-Item, Avg-Last-10, Content (mean embedding), CF-ItemKNN, MIND, ComiRec, Popularity and Random, plus SASRec and GRU4Rec as library-implemented sequential models (Sect. 5.6). Recency-Weighted is the critical one: it is Essence with the clustering step deleted, so the Essence–Recency-Weighted contrast isolates clustering exactly.

Table 2. Recall@10 at adopted K. Baselines at the better of published and tuned. Best per dataset in bold.

System	Last.fm-1K	Amazon Books	MovieLens-25M
Essence	0.0281	0.0246	0.0055
Recency-Weighted	**0.0304**	0.0260	0.0076
Last-Item	0.0265	**0.0311**	0.0091
Avg-Last-10	0.0252	0.0235	0.0064
Content (Avg Emb)	0.0038	0.0173	0.0043
CF (ItemKNN)	0.0018	0.0031	**0.0850**
MIND	0.0082	0.0072	0.0423
ComiRec (SA)	0.0243	0.0056	0.0706
Popularity	0.0023	0.0021	0.0517
Random	0.0001	0.0001	0.0012

Conservative baseline convention. Each baseline is reported at the *better of its published and validation-tuned configuration on test*. This is deliberately not the textbook protocol. We adopt it because strict validation-selection would in one case have reported a ComiRec configuration 15.8% worse on test than one we had already run, turning a tuning artifact into an apparent win for our own system. The convention favours the baselines and can only weaken our results. It costs us one previously significant comparison, reported in Sect. 5.1.

Statistics. Paired bootstrap over users, 10,000 resamples, two-sided p counting exact ties in both tails, Benjamini–Hochberg FDR within family, Cohen's d for paired differences. For every null we also report the minimum detectable effect at 80% power, estimated from the bootstrap standard error (Sect. 5.3).

5 Results

5.1 Domain dependence

Table 2 shows the pattern: Essence ranks second and third of ten on Last.fm-1K and Amazon Books, and eighth of ten on MovieLens-25M, where CF-ItemKNN alone beats it by more than an order of magnitude. Table 3 shows the ordering survives every cutoff and NDCG@10.

The effect is not a property of clustering. Recency-Weighted—identical to Essence but with no clustering at all—exhibits the same profile, ranking first on Last.fm-1K and collapsing on MovieLens-25M. What the data separate is architectures that pool across users from architectures that do not.

What the conservative convention costs. Essence significantly outperforms MIND on both sparse datasets and ComiRec on Amazon Books, but is *indistinguishable from a properly tuned ComiRec on Last.fm-1K* ($d = 0.065$, $q = 0.54$): tuning closed that gap from 2.05× to 1.15×, and our earlier claim rested on a ComiRec

Table 3. Essence holds the same rank under every metric on both datasets where the full set was computed; MIND and ComiRec are re-scored from the same ranked lists. Two disclosures, neither changing any ordering: the Amazon ComiRec row uses its tuned configuration, 16% below the conservative value in Table 2, and the MovieLens rows are re-runs at lr $= 10^{-3}$, $K = 4$, within -0.8% and $+11.2\%$ of published.

	Amazon Books			MovieLens-25M		
System	R@5	R@10	NDCG@10	R@5	R@10	NDCG@10
Last-Item	0.0238	0.0311	0.0364	0.0076	0.0091	0.0151
Recency-Weighted	0.0170	0.0260	0.0273	0.0044	0.0076	0.0111
Essence	0.0162	0.0246	0.0262	0.0035	0.0055	0.0078
Avg-Last-10	0.0154	0.0235	0.0243	0.0042	0.0064	0.0101
Content (Avg Emb)	0.0111	0.0173	0.0174	0.0019	0.0043	0.0052
MIND	0.0043	0.0072	0.0071	0.0248	0.0470	0.0546
ComiRec (SA)	0.0027	0.0047	0.0043	0.0399	0.0700	0.0899
CF (ItemKNN)	0.0019	0.0031	0.0032	0.0469	0.0850	0.1093
Popularity	0.0014	0.0021	0.0021	0.0307	0.0517	0.0712
Random	0.0000	0.0001	0.0001	0.0009	0.0012	0.0021

denied a fair learning rate. The comparison is also underpowered (detectable effect 0.0159 against an observed 0.0038), so the two are *not shown to differ* at $n = 99$—not equal. The defensible claim is thus narrower than before: peer-isolated retrieval is *competitive with* tuned multi-interest recommenders where co-occurrence is sparse, and far behind where it is dense.

5.2 The clustering null

The sharpest version of the test restricts attention to the users whose taste is most clearly multi-modal, stratifying by silhouette score: if clustering helps anyone, it should help them. At validation-selected K, with tie-aware p and no rows excluded, clustering is significant in none of those 18 stratified comparisons against recency-weighted retrieval. Our earlier work reported "one of sixteen", where the sixteen were a hand-curated subset—formed, we later found, by dropping exactly the degenerate cases the defect in Sect. 5.13 produced—and the one significant row was itself an artifact of that defect.

Table 4 states the null at full scope.

Across all fourteen comparisons—three datasets, two encoders, two activity cohorts, two split protocols—exactly one favours clustering and one favours recency-weighting, both marginal.

The exception, reported and not claimed. Under the global cutoff, clustering beats recency-weighting on long-tail recall at $q = 0.0496$. It is not an artifact of fallback users: restricted to the 357 users who genuinely cluster, it survives at $p = 0.0348$, $d = 0.096$. We nonetheless decline to claim it. The observed effect is $0.64\times$ its own 80%-power detection threshold, placing it in the regime where

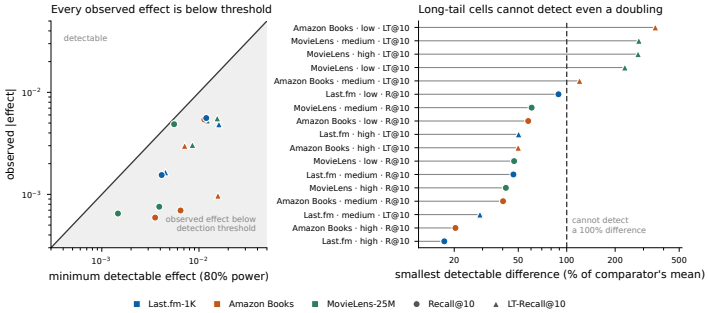


Fig. 2. Observed effect against minimum detectable effect at 80% power for every null comparison. Points below the diagonal are effects the test could not have detected. Ranking-metric nulls are adequately powered; several long-tail cells are not and are not claimed.

Table 4. Every Essence versus Recency-Weighted comparison in this study.

Robustness check	Configuration	Null holds?
Main analysis	3 datasets, adopted K	yes
Second sentence encoder	all-mpnet-base-v2, 2 datasets	yes (0 of 8 favour clustering)
Heavy users	330 interactions/user, $K = 60$	yes ($q = 0.31$, powered to 21.6%)
Global temporal cutoff	single wall-clock split	Recall@10 yes; **one exception**
Silhouette-stratified	most multi-modal users	yes

significant findings are most inflated; it is one comparison out of fourteen, where 0.7 false positives are expected at $\alpha = 0.05$; and $d = 0.09$ is negligible beside the effects the rest of the paper turns on.

5.3 Statistical power

A null is only informative if the test could have detected an effect. Of 18 null comparisons, 17 are informative and none reach their own minimum detectable effect (Fig. 2). The comparisons split cleanly into two regimes. Nulls on ranking metrics are adequately powered and carry content. Several long-tail cells cannot detect even a doubling of the comparator, and we do not assert those as nulls—a correction to our earlier work, which did.

5.4 Heavy users

All three datasets bound user activity from above, which raises an obvious objection: the excluded heavy users are the most plausibly multi-modal, so the

clustering null may have been established on the population least likely to exhibit the effect it tests.

We replicated on Amazon Books users with 201–1000 interactions (1,757 users, 330 interactions each against ∼29, catalogue 213,221 items, identical text coverage), re-selecting $K = 60$ for that cohort so clustering received its best configuration. The null holds: Recall@10 difference −0.000288 ($d = -0.027$, $q = 0.31$) with the test powered to detect a 21.6% relative difference. Clustering does not help the users it was supposed to help.

Two secondary observations. Essence slips from third to fourth of the eight comparable systems, and CF-ItemKNN degrades by only 22% where content methods lose 86–93%: heavy users supply the co-occurrence signal CF needs. This is the mechanism of Sect. 5.9 appearing a second time, in a different dataset and under a different manipulation — raising the activity floor rather than lifting the ceiling — and it is the reason we frame the first finding as population density rather than domain.

5.5 Encoder sensitivity

Replacing all-MiniLM-L6-v2 (384-dim) with all-mpnet-base-v2 (768-dim) leaves conclusions intact. Essence improves slightly (Amazon Books 0.0246 to 0.0254, rank three to two) and only one of eighteen significance verdicts changes. Selected K is identical under both encoders. As a correctness check, the five systems that do not consume item embeddings return bit-identical results while exactly the five that do consume them move.

5.6 Sequential baselines

SASRec and GRU4Rec were run through RecBole with learning rates tuned on held-out validation, and scored with our own metric implementations from the library's ranked lists. Both fail on these datasets: Essence leads by 20.9× and 23.3× on Last.fm-1K and by 174.6× and 59.3× on Amazon Books. Even taking each model's best test configuration—a deliberately unfair comparison that favours the baselines—the gap is 11.9× to 43.6×.

The cause is structural rather than a tuning failure. Amazon Books is 0.063% dense with 32 training items per user, so there is very little transition structure to learn. This is the same weak-collaborative-signal condition our thesis concerns, and the result is therefore consistent with it rather than incidental. It is emphatically *not* evidence that Essence outperforms SASRec in general; on denser data we would expect the ordering to reverse, as it already does for the other collaborative baselines on MovieLens-25M.

5.7 Temporal leakage

Per-user chronological splitting allows one user's training window to overlap another's test window, which can favour collaborative methods. Under a single

Table 5. The between-dataset difference as an interaction, not a ranking. Δd is the change in Essence's effect size when moving from a sparse-co-occurrence dataset to MovieLens-25M. Every contrast is significant after BH correction and every one is a sign flip: Essence ahead becomes Essence behind. These hold for the datasets *as configured*; Sect. 5.9 shows Last.fm-1K's position is set by its user cap, not by its domain.

Baseline	Sparse-density set	d (weak)	d (MovieLens)	Δd	95% CI
CF (ItemKNN)	Last.fm-1K	+0.28	−0.55	0.83	[0.76, 0.95]
CF (ItemKNN)	Amazon Books	+0.25	−0.55	0.80	[0.76, 0.84]
MIND	Last.fm-1K	+0.24	−0.38	0.62	[0.54, 0.76]
MIND	Amazon Books	+0.21	−0.38	0.58	[0.54, 0.63]
ComiRec (SA)	Last.fm-1K	+0.07	−0.52	0.58	[0.33, 0.72]
ComiRec (SA)	Amazon Books	+0.22	−0.52	0.74	[0.70, 0.78]
Popularity	Last.fm-1K	+0.27	−0.44	0.71	[0.63, 0.82]
Popularity	Amazon Books	+0.26	−0.44	0.69	[0.65, 0.74]

global cutoff every system degrades, but unevenly: CF-ItemKNN loses 86% where content and recency methods lose 37–45%, and Essence's rank improves from third to second of eight. The leakage was inflating the collaborative baselines, so our main protocol is the conservative choice for this paper's central comparison.

5.8 Is the domain difference real? A formal test

Ranks describe the pattern; they do not test it. We therefore test the interaction directly. For each collaborative baseline we compute the paired effect size (Cohen's d) of Essence against it within each dataset, bootstrap each dataset's d independently (10,000 resamples), and take the distribution of the difference. d is scale-free, which matters because absolute recall differs by an order of magnitude across these datasets.

All 8 contrasts are significant after BH correction, all 8 show a sign flip, and no confidence interval contains zero; the smallest lower bound is 0.33. Δd ranges from 0.58 to 0.83, which are large effects. The difference between datasets is therefore a tested claim rather than a described pattern. Sect. 5.9 then shows that a dataset's position on this axis is set by its cohort rather than its domain: Last.fm-1K's d against CF-ItemKNN changes sign when its user cap is lifted, so the contrasts above hold for these datasets *as configured* and should not be read as properties of music, book retail or film.

5.9 Cohort density, not domain, drives the verdict

Sect. 5.8 establishes that Essence's standing differs between datasets. It does not establish *why*, and the natural reading — that music and book retail simply carry weaker collaborative signal than film — turns out to be wrong. The Last.fm-1K cohort is capped at 500 distinct tracks per user, which admits 99 of the 975 users

Table 6. The same dataset, the same domain, the same code and seed. Raising the user cap raises cross-user co-occurrence and reverses the verdict against item-based collaborative filtering. Mean training history rises from 215 to 783 items per user and the candidate catalogue from 18,679 to 158,074 items, so absolute recall falls for every system; only within-row comparisons are meaningful.

Cohort (tracks/user)	Users	Items w/ ≥ 2 users	Essence	CF-ItemKNN	d
50–500 (published)	99	**10.3%**	0.0281	0.0018	**+0.28**
50–2,000	374	**28.4%**	0.0026	0.0055	**−0.16**

clearing the minimum. Lifting that cap to 2,000 admits 374, and reverses the result.

Both verdicts are significant ($q < 0.0001$ and $q = 0.014$). On the expanded cohort CF-ItemKNN is the best of the eight systems and Essence falls from second to fifth. The mediating quantity is in the fifth column: item-based collaborative filtering requires cross-user co-occurrence, and the expanded cohort supplies nearly three times as much of it.

This reframes the first finding rather than overturning it. Signal strength is a property of the *evaluated population*, not the content domain, and the three-dataset pattern is three points on that one axis — which is why MovieLens-25M, the densest, is where peer-isolated retrieval fails worst. It is also actionable: measure co-occurrence density in your own user base rather than reason from your product category.

A confound we cannot remove. Lifting the cap raises the user count (99 to 374) and the history length (215 to 783 items/user) together. Both raise density, so both act through the same mediator, but this design cannot separate them: isolating user count would need 374 users with 50–500 interactions and only 99 exist. We claim density as the mediator, not which of its causes dominates. MIND and ComiRec are omitted from Table 6: trained on the 99-user sample, they cover only 26.5% of the expanded cohort, and mixing a 99-user evaluation into a 374-user ranking is the error this paper elsewhere reports catching. The clustering null is unaffected and moves marginally further against clustering (LT-Recall@10 $q = 0.031$).

5.10 Study-wide multiplicity

FDR above is controlled within each family, which is conventional but invites a fair question: how many hypotheses does this study test in total? The answer is 136. Applying Benjamini–Hochberg across all 136 at once takes the significant count from 59 to 56 and changes exactly two rows, both against clustering.

First, the one exception in Sect. 5.7 — the single comparison in this study where clustering beat recency-weighting — is not significant study-wide ($q = 0.075$). We had already declined to claim it on power grounds; the stricter cor-

rection independently agrees. Under it, *no comparison anywhere in this study shows clustering significantly outperforming recency-weighted retrieval.*

Second, one comparison becomes significant that was not: in the silhouette-stratified test the *high*-silhouette stratum — the users whose histories separate most cleanly into distinct interest groups, which is precisely the population the multi-interest premise concerns — shows clustering performing significantly *worse* than recency-weighting ($q = 0.033$). This is not a paradox: BH adapts to the distribution of p-values it is given, and a family of 18 mostly-null tests is a less favourable reference than the full study. It is the strongest form the negative result takes. On the users where interest clustering should help most, it measurably hurts.

We report within-family FDR as the primary correction and state the study-wide result alongside it. Reporting only the former would preserve an exception we do not believe; reporting only the latter would overstate the certainty of a post-hoc choice.

5.11 Seed stability

Two seeds were fixed and never varied: the per-user K-means initialisation and the random user sample. Both were re-run on Amazon Books at the adopted K. The clustering null holds in all five configurations, and significance counts are identical across the three clustering seeds and across the two user samples; no verdict depends on either seed.

Worth stating explicitly, since the opposite would be a fair criticism: the published seed is mildly *unfavourable* to Essence. At `random_state=42` the comparison against Recency-Weighted gives $q = 0.52$; the two alternative seeds give $q = 0.61$ and $q = 0.90$, i.e. further from significance, and Essence scores 8.0% higher under them on average. The published configuration does not flatter the method.

5.12 The identity shortcut, at the adopted K

Our earlier work reported that roughly 20% of Essence's hits share an artist with the user's active cluster, and that excluding same-artist candidates collapses performance by 85–90%. That analysis was run at $K = 3$, a value this paper does not adopt — an inconsistency we found while re-verifying it. At $K = 3$ the figures reproduce exactly (20.6% shortcut share, 87.9% collapse). At the adopted $K = 10$ the dependence is substantially larger: **32.9% of hits are same-artist and removing those candidates costs 95.3% of performance**.

More clusters means tighter, more artist-homogeneous clusters, so nearest-centroid retrieval lands closer to artists already inside the active cluster. The shortcut is a property of the retrieval geometry and it grows with K. Restating it at the adopted K therefore sharpens rather than softens the limitation: a large part of what this architecture recovers is identity matching, not taste modelling.

Cold start. At the adopted K, content-only retrieval beats CF-ItemKNN in 5 of 6 popularity-decile comparisons at the two coldest deciles across the three datasets, the exception being MovieLens's second decile where CF has begun to acquire signal. This is a generic consequence of not requiring co-occurrence, not a property of Essence, and it is the same mechanism as Sect. 5.9 at the item level rather than the population level.

5.13 A statistical defect in our earlier work

The two-sided bootstrap p-value was computed as $\min(1, 2\min(f, 1-f))$ with $f = \text{mean}(\delta > 0)$ over resampled differences δ. This counts every resample with $\delta = 0$ as evidence against the null. On long-tail recall, where most users score exactly zero, the resample distribution has large mass at zero, so a one-sided-with-ties difference was reported as $p = 0$—including cases in which both arms were identically zero for every user and no difference existed to detect.

We fixed this in all call sites by counting ties in both tails. The correction is consequential and moves against us: the clustering-value test falls from seven of eighteen significant to none of 18. We verified the fix is surgical rather than wholesale—one dataset's corrected output is byte-identical to its published counterpart.

6 Discussion and Conclusion

The practical question has a conditional answer: a user's own history is enough when cross-user co-occurrence is thin and not enough when it is dense. That boundary belongs to the *cohort*, not the domain — the same music data sits on either side of it depending on the user cap — so the diagnostic for a practitioner under an on-device or siloed-data constraint is not which algorithm to pick, but whether their user base supplies dense co-occurrence.

The clustering null carries a narrower lesson. The multi-interest premise—that users hold several tastes one vector cannot capture—may well be true; our data show only that recovering them by unsupervised partition of one user's embedded history yields nothing a recency-weighted query does not already have.

Conclusion. Peer-isolated retrieval is a viable design where co-occurrence is sparse and a poor one where it is dense, and that boundary is set by the cohort rather than by the domain. Within the design, per-user interest clustering—the component we originally proposed as the contribution—adds nothing measurable over simple recency weighting across every robustness dimension we tested.

Limitations. Item text quality bounds every content method here and we did not vary it. The long-tail nulls are underpowered and we do not claim them. The main Last.fm-1K analysis has 99 users, which limits power on that dataset; Sect. 5.9 reports the expanded cohort. Sequential baselines on Amazon Books use a split agreeing with ours on 92.8% of test items rather than exactly.

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